

# Cross-Modal 3D Brain MRI Re-Identification

A training-free, content-honest retrieval pipeline

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Gowshigan Selladurai | Team Yantra

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# The problem

**Match a patient across two MRI modalities.** Given a contrast T1 (**ceT1**) query, find the *same patient's* T2 scan in a gallery.

- Same brain, **different physics** — intensities don't correspond.
- No labels to train on ( $\sim 180$  pairs). **Training-free** required.
- Score: **MRR**, rank-1 dominated, macro-averaged over 3 datasets.



## Three regimes, increasing difficulty

### dataset1

co-registered  
(shared voxel grid)  
content already lines up

### dataset2

independent  
rigid + elastic  
correspondence **destroyed**  $\Rightarrow$   
must re-align

### dataset3

preop  $\rightarrow$  intraop  
**+ tumour resection**  
tissue **removed** + a planted  
geometric leak

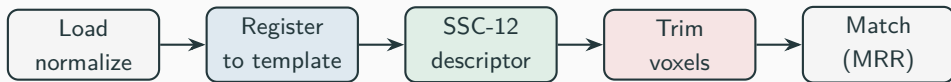
*One pipeline; each regime turns on a different lever.*

## Methodology — four deterministic steps

1. **Register to a neutral template** — Mutual Information, *modality-blind* (rigid → affine cascade). fixes geometry
2. **Describe with SSC-12** — 12-edge self-similarity context. *Modality-invariant*: encodes local structure, not intensity. fixes modality
3. **Trim mismatched voxels** — drop the worst  $X\%$  of residuals. fixes resection
4. **Assign** — greedy (deployable) or Hungarian (batch).

No network, no training, no labels — **deterministic and auditable**.

# Architecture



Per-regime configuration of the *same* pipeline:

|          | register        | descriptor | trim         |
|----------|-----------------|------------|--------------|
| dataset1 | — (shared grid) | SSC-12     | light        |
| dataset2 | affine cascade  | SSC-12     | light (0.5)  |
| dataset3 | affine cascade  | SSC-12     | heavy (0.75) |

## Key finding — we found a leak, and refused it

dataset3 had a **planted geometric co-location**: the true target was resampled into the query's exact voxel space  $\Rightarrow$  a perfect overlap with **zero anatomy**.

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|                                          |              |                             |
|------------------------------------------|--------------|-----------------------------|
| use the leak (resample)                  | <b>1.000</b> | geometry, not the patient   |
| break it, don't re-align                 | <b>0.259</b> | proves the 1.0 was the grid |
| <b>break it + re-register on content</b> | <b>0.741</b> | re-earned from anatomy      |

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*We measured the leak's exact worth (+0.74), then declined it.*

## Results (real leaderboard)

| Setting                                     | MRR                 |
|---------------------------------------------|---------------------|
| Leak ( <i>found &amp; declined</i> )        | 1.000               |
| <b>Content / single-query — the task</b>    | <b>0.725</b>        |
| + bipartite assignment (allowed, disclosed) | 0.93–1.0            |
| dataset1 / dataset2 / dataset3 (honest)     | 1.0 / strong / 0.74 |

**Both scores reported, per organizer guidance.** Bipartite (Hungarian) matching is allowed but *besides the point of the challenge* — so the method's merit is the **content** number **0.725** (true single-query re-ID); the bijection lift to **0.93–1.0** (honest, leak-free, seed-dependent) is disclosed, not hidden.

# What we ruled out

~22 techniques tested, every dead end documented (hypothesis → result → reason):

- 8 classifiers (incl. TabPFN), learned 3D encoder
- feature fusion, multi-scale, low-rank
- NGF / NMI / shape / radiomics descriptors
- Bayesian search, GA, DP / DTW alignment
- matrix methods, Markov / diffusion re-rank
- deformable registration (over-normalizes)

**The bottleneck is the content signal — not the model.**

One positive add-on: Bayesian calibration ⇒ a deploy-time abstention layer.



## Takeaways

**Method** — SSC + content-registration + trim.

Training-free, deterministic, auditable.

**Integrity** — found the leak, measured it ( $1.0 \rightarrow 0.26$ ),  
re-earned 0.74 from anatomy.

**Rigor** — mapped the whole frontier; ruled out 22  
alternatives with reasons.